CUMULATIVE FINAL · 35 QUESTIONS

# Pre-Calculus

A cumulative assessment built from your class's actual exam data — weighted by what you missed most across the four units. 28 multiple-choice questions, plus 2 free-response problems with 7 typed sub-parts.

4 Units 28 Multiple Choice (1-28) 2 FRQ · 7 Typed Inputs (29-30)

6 Options Per MC Auto-Graded

📋 DIRECTIONS

Q1-Q28: Select the single best answer (A-F). Your selection highlights but does not lock — you can change it anytime before submitting. Q29-Q30: Type your numeric or symbolic answer in each input field. After clicking Submit Exam at the bottom, every question locks, your score appears, and an explanation reveals on each item.

## UNIT 01 Polynomial & Rational Functions

### 01

1A · TOPIC 1.2 AVG RATE OF CHANGE

A function $f$ satisfies $f(2) = 5$ and $f(7) = 20$. What is the average rate of change of $f$ on the interval $[2, 7]$?

A $-3$

B $\frac{1}{3}$

C 5

D $\frac{25}{9}$

E 3

F 15

02 1A · TOPIC 1.3 CONCAVITY

A continuous function f is increasing on [0, 10]. On the interval (0, 5), f is concave down; on the interval (5, 10), f is concave up. What feature does the graph have at x = 5?

A Local maximum

B Local minimum

C Point of inflection

D Vertical asymptote

E Horizontal asymptote

F Removable discontinuity

## 03

1B · TOPIC 1.4 🔥 BIGGEST MISS

The graph of a polynomial $f$ is increasing on the entire interval $(1, 7)$. Between $x = 3$ and $x = 5$ the curve has an obvious S-bend where the concavity switches from concave up to concave down at $x = 4$. The point at $x = 4$ is best classified as:

A Local maximum (the curve reaches a peak there)

B Local minimum (the curve reaches a valley there)

C Point of inflection (concavity changes; direction does not)

D Global maximum on the interval

E Both a local max and a point of inflection

F A vertical asymptote

## 04

1B · TOPIC 1.6 END BEHAVIOR

For the polynomial $f(x) = -2x^4 + 5x^3 - x + 9$, which statement correctly describes the end behavior?

A As $x \to \pm\infty$, $f(x) \to +\infty$

B As $x \to -\infty$, $f(x) \to -\infty$ and as $x \to +\infty$, $f(x) \to +\infty$

C As $x \to -\infty$, $f(x) \to +\infty$ and as $x \to +\infty$, $f(x) \to -\infty$

D As $x \to \pm\infty$, $f(x) \to -\infty$

E As $x \to \pm\infty$, $f(x) \to 0$

F End behavior cannot be determined without more information

## 05

1B · TOPIC 1.5 MULTIPLICITY

For the polynomial $f(x) = 3(x + 2)^2(x - 1)(x - 4)^3$, identify the behavior of the graph at each x-intercept.

A $x = -2$ S-crosses, $x = 1$ bounces, $x = 4$ crosses cleanly

B $x = -2$ crosses cleanly, $x = 1$ bounces, $x = 4$ S-crosses

C All three intercepts cross cleanly

D $x = -2$ bounces, $x = 1$ S-crosses, $x = 4$ crosses cleanly

E $x = -2$ bounces, $x = 1$ crosses cleanly, $x = 4$ bounces

F $x = -2$ bounces, $x = 1$ crosses cleanly, $x = 4$ S-crosses

## 06

1C · TOPIC 1.7 HOLES VS ASYMPTOTES

For the rational function $r(x) = (x - 3)(x + 1) / [(x - 3)(x + 5)]$, identify all holes and vertical asymptotes.

A Vertical asymptotes at $x = 3$ and $x = -5$; hole at $x = -1$

B Vertical asymptote at $x = -5$; hole at $x = 3$

C Holes at $x = 3$ and at $x = -5$

D Vertical asymptotes at $x = 3$ and $x = -5$ (no holes)

E Vertical asymptote at $x = 3$; hole at $x = -5$

F No holes; vertical asymptote at $x = -1$ only

07 1C · TOPIC 1.8 ASYMPTOTES

Which statement best describes the long-run behavior of $f(x) = (4x^3 + 1) / (x^2 + 5)$ ?

A Horizontal asymptote at $y = 0$

B Horizontal asymptote at $y = 4$

C Horizontal asymptote at $y = 1/5$

D Horizontal asymptote at $y = 4x$

E No horizontal asymptote — the function has a slant (oblique) asymptote

F Horizontal asymptote at $y = \infty$

UNIT 02 Exponential & Logarithmic Functi...

08 2A · TOPIC 2.2 GEOMETRIC SEQ

A geometric sequence has first term $a_1 = 6$ and common ratio $r = 2$. What is the value of the 6th term, $a_6$?

A 12

B 36

C 96

D 192

E 384

F 64

09 2A · TOPIC 2.4 FRACTIONAL EXPONENTS

Simplify the expression 27^(2/3) .

A 18

B 6

C 81

D $\frac{27}{2}$

E 3

F 9

**10** 2B · TOPIC 2.12 LOG PROPERTIES

Which expression is equivalent to $\log(x^2\sqrt{y} / z^3)$ , where the $\sqrt{}$ denotes square root and all variables are positive?

A $2 \log x + (1/2) \log y - 3 \log z$

B $2 \log x \cdot (1/2) \log y / 3 \log z$

C $\log(2x) + \log(\sqrt{y}) - \log(3z)$

D $2 \log x + \log(\frac{1}{2}) \cdot \log y - 3 \log z$

E $2 \log x + (1/2) \log y + 3 \log z$

F $\log(2x + \sqrt{y} - 3z)$

**11** 2B · TOPIC 2.13 LOG EQUATION

Solve for x: $\log_4(x) + \log_4(x - 6) = 2$ .

A $x = 2$

B $x = 4$

C $x = 6$

D $x = 8$

E $x = 16$

F $x = 22$

12 2B · TOPIC 2.13 🔥 BIGGEST MISS · 5/10 WRONG

What is the x-intercept of the function $f(x) = \log_3(x - 1) - 2$ ?

A $x = 1$

B $x = 10$

C $x = 4$

D $x = 9$

E $x = 2$

F $x = 27$

13 2B · TOPIC 2.15 🔥 SEMI-LOG · 4/10 WRONG

When the points $(x, y)$ of a data set are graphed as $(x, \log_{10} y)$ on a semi-log plot, the result is a line with slope 0.5 and y-intercept 1. Which formula best models the original function $y(x)$?

A $y = 0.5x + 1$ (simple linear)

B $y = \log(0.5x + 1)$

C $y = 0.5 \cdot 10^x$

D $y = x^{(0.5)} + 10$

E $y = 10 \cdot 10^{(0.5x)}$

F $y = e^{(0.5x + 1)}$

14 2C · TOPIC 2.8 🔥 HIGHEST MISS IN UNIT 2

The graph of $f$ passes through the points (1, 4), (3, 7), and (5, 12). Through which set of points must the graph of $f^{-1}$ (the inverse of $f$) pass?

A (1, ¼), (3, 1/7), (5, 1/12)

B (-1, -4), (-3, -7), (-5, -12)

C (1, 4), (3, 7), (5, 12)

D (4, -1), (7, -3), (12, -5)

E (¼, 1), (1/7, 3), (1/12, 5)

F (4, 1), (7, 3), (12, 5)

UNIT 03 Trigonometric & Polar Functions

**15** 3A · TOPIC 3.2 🔥 RADIAN DEF · 5/10 WRONG

Which statement is the correct **definition** of one radian?

A An angle equal to $180/\pi$ degrees exactly

B The central angle of a circle that subtends an arc whose length equals the radius of the circle

C An angle whose terminal side passes through the point (1, 0)

D An angle of measure $360°/\pi$

E Half of one full revolution around a circle

F The angle whose sine equals 1

**16** 3A · TOPIC 3.4 EXACT VALUES

Find the exact value of $\cos(7\pi/6)$ .

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

E $\frac{\sqrt{2}}{2}$

F $-\frac{\sqrt{2}}{2}$

## 17

3B · TOPIC 3.6

A·SIN(BX+C)+D

A sinusoidal function is given by $f(x) = 3\cdot\sin(\pi x/4) + 5$ . What is its period?

A 8

B 4

C $\pi/4$

D $2\pi$

E $8\pi$

F $\pi$

## 18

3B · TOPIC 3.7

SINUSOIDAL MODELING

A Ferris wheel has a diameter of 60 m and its center is 35 m above the ground. It completes one full revolution every 4 minutes. A rider boards at the lowest point at time t = 0. Which equation models the rider's height h(t) above the ground (in meters)?

A $h(t) = 30\cdot\sin(\pi t/2) + 35$

B $h(t) = 60 \cdot \cos(\pi t/4) + 35$

C $h(t) = 30 \cdot \cos(\pi t/2) + 35$

D $h(t) = -30 \cdot \cos(\pi t/2) + 35$

E $h(t) = 35 \cdot \sin(\pi t/2) + 30$

F $h(t) = -60 \cdot \sin(\pi t/4) + 35$

## 19

3B · TOPIC 3.10 TRIG EQUATIONS

Find *all* solutions to $2 \cdot \sin(x) - 1 = 0$ on the interval $[0, 2\pi)$.

A $x = \pi/6$ only

B $x = \pi/3$ only

C $x = \pi/3$ and $x = 2\pi/3$

D $x = \pi/4$ and $x = 3\pi/4$

E $x = \pi/6$ and $x = 5\pi/6$

F $x = \pi/6$ and $x = 7\pi/6$

## 20

3C · TOPIC 3.14 🔥 POLAR GRAPHS · 6/10 WRONG

Which polar equation produces a **four-petaled rose**?

A $r = 3\cos(2\theta)$

B $r = 4\sin(\theta)$

C $r = 2 + \cos(\theta)$

D $r = 2\cos(3\theta)$

E $r = 1 + 2\cos(\theta)$

F $r = 4$

21 3C · TOPIC 3.12 DOUBLE-ANGLE

Which expression is equivalent to $\sin(2\theta)$ for all real $\theta$?

A $\sin(\theta) + \cos(\theta)$

B $2\sin(\theta)$

C $2\sin(\theta) \cdot \cos(\theta)$

D $1 - 2\sin^2(\theta)$

E $\cos(\theta) \cdot \sin(\theta)$

F $\sin^2(\theta) - \cos^2(\theta)$

UNIT 04 Parametric, Implicit Relations & C...

## 22

4A · TOPIC 4.1 FIND PARAMETER

A parametric curve is defined by $x(t) = t^2 - 1$ and $y(t) = 2t + 3$. For which value of t is the point on the curve equal to (3, 7)?

A $t = 2$

B $t = 1$

C $t = -2$

D $t = 3$

E Both $t = 2$ and $t = -2$

F $t = 4$

## 23

4A · TOPIC 4.3 RATES OF CHANGE

A parametric curve has $x(t) = 3t^2$ and $y(t) = t^3$. What is the average rate of change of y with respect to x on the interval $t \in [1, 2]$?

A 3

B $\frac{7}{3}$

C $\frac{7}{9}$

D $9$

E $\frac{9}{7}$

F $\frac{1}{3}$

## 24

4B · TOPIC 4.4 PARAMETRIC CIRCLE

Which set of parametric equations describes a complete **circle** of radius 5 centered at (2, −3)?

A $x = 5 + 2\cos(t), \quad y = 5 - 3\sin(t), \quad 0 \le t < 2\pi$

B $x = 2\cos(t), \quad y = -3\sin(t), \quad 0 \le t < 2\pi$

C $x = 2 + 5t, \quad y = -3 + 5t, \quad 0 \le t < 2\pi$

D $x = 5\cos(t) - 2, \quad y = 5\sin(t) + 3, \quad 0 \le t < 2\pi$

E $x = 2 + 5\cos(t), \quad y = -3 + 5\sin(t), \quad 0 \le t < 2\pi$

F $x = 2 + 5\cos(t), \quad y = 3 + 5\sin(t), \quad 0 \le t < 2\pi$

## 25

4B · TOPIC 4.5 IMPLICIT · INTERSECTION

How many points of intersection are there between the implicit relation $x^2 + y^2 = 25$ and the line $y = x + 1$ ?

A 0

B 1

C 3

D 4

E Infinitely many

F 2

26 4C · TOPIC 4.6 ELLIPSE STANDARD FORM

An ellipse is centered at (1, −2). Its major axis is horizontal with total length 10, and its minor axis is vertical with total length 6. Which equation represents this ellipse in standard form?

A $(x + 1)^2/25 + (y - 2)^2/9 = 1$

B $(x - 1)^2/25 + (y + 2)^2/9 = 1$

C $(x - 1)^2/100 + (y + 2)^2/36 = 1$

D $(x - 1)^2/9 + (y + 2)^2/25 = 1$

E $(x - 1)^2/10 + (y + 2)^2/6 = 1$

F $(x - 1)^2/5 + (y + 2)^2/3 = 1$

## 27

4C · TOPIC 4.6 🔥 HYPERBOLA · Q20 MISS

For the hyperbola $(x - 2)^2/16 - (y + 1)^2/9 = 1$, what are the equations of the two asymptotes?

A $y = \pm(4/3)(x - 2) - 1$

B $y = \pm(3/4)(x + 2) + 1$

C $y = \pm(16/9)(x - 2) - 1$

D $y = \pm 3(x - 2) - 1$

E $y = (3/4)\,x$ and $y = -(3/4)\,x$

F $y = \pm(3/4)(x - 2) - 1$

## 28

4C · TOPIC 4.7 PARAMETRIC CONIC

The parametric curve $x = 4\cos(t)$, $y = 3\sin(t)$, $0 \le t < 2\pi$, traces:

A A circle of radius 4

B An ellipse with semi-major axis 4 (horizontal) and semi-minor axis 3 (vertical)

C A circle of radius 3

D A hyperbola

**E** A line through the origin

**F** A parabola

FREE RESPONSE

## Q29–Q30 · 7 Typed Sub-Parts

**29** 3B · TOPIC 3.6 / 3.10 SINUSOIDAL MODELING · 4 PARTS

CONTEXT

The water level in a tidal harbor is modeled by a sinusoidal function of time. The level reaches its **maximum of 22 ft at t = 6 hours** and its **minimum of 6 ft at t = 18 hours**, where t is measured in hours after midnight. Use this to answer parts (a)–(d). Type your answers in the boxes below.

(a) Find the **amplitude A** and the **midline (vertical shift) D** of the sinusoidal model.

A = amplitude D = midline

(b) Determine the **period** (in hours) and the value of **B** in the form cos(B(t − C)) . Use π in your answer for B (e.g. pi/12 ).

Period = hours B = e.g. pi/12

(c) Write the **full sinusoidal equation** h(t) using cosine, with the maximum at t = 6 (so the phase shift puts the peak there).

h(t) = e.g. 8*cos((pi/12)*(t-6)) + 14

(d) Evaluate $h(12)$. Give the water level (in feet) at $t = 12$ hours (noon).

h(12) = ft

30

4C · TOPIC 4.7 / 4.10

HYPERBOLA → PARAMETRIC · 3 PARTS

CONTEXT

Consider the hyperbola defined by $(x - 1)^2 / 9 - (y + 2)^2 / 16 = 1$. Use this for parts (a)-(c). Type your answers in the boxes below.

(a) Identify the **center** as an ordered pair, and the two **vertices**. Use the format (x,y).

Center = (h,k) Vertex 1 = (x,y)

Vertex 2 = (x,y)

(b) Write the equations of the two **asymptotes** in slope-intercept form (use y = ). Order doesn't matter.

Asym 1: y = ... Asym 2: y = ...

(c) Give a **parametric form** using sec(t) and tan(t), then evaluate the **point at t = 0** as an ordered pair.

x(t) = e.g. 1 + 3*sec(t) y(t) = e.g. -2 + 4*tan(t)

Point at t = 0: (x,y)

END OF EXAM

/ 35 ANSWERED

SUBMIT EXAM → RESET ALL

After submitting, every question locks · scores reveal · explanations open. Submit when you're ready.

Stratosyn · 2026

Built for Room 501 · ALA Queen Creek